

Key Responsibilities

1. Support the design and development of power electronic circuits and subsystems for EV applications.
2. Assist in schematic design, PCB layout, and circuit simulation using tools like LTspice, PSpice, or equivalent.
3. Participate in hardware bring-up, prototype debugging, and validation activities.
4. Capture, analyze, and troubleshoot electrical waveforms using oscilloscopes, power analyzers, and other lab instruments.
5. Support component selection, BOM preparation, and alternate component evaluation.
6. Assist in functional, performance, thermal, endurance, reliability, and safety testing of hardware systems.
7. Set up and operate electrical, thermal, and mechanical test benches, including support for EVT, DVT, and PVT validation phases.
8. Contribute to EMI/EMC pre-compliance, electrical safety testing, root-cause analysis, and troubleshooting during product development.
9. Prepare technical reports, test documentation, and collaborate with hardware, firmware, quality, manufacturing, and supplier teams

Required Qualifications

1. B.Tech/B.E. in Electrical Engineering, Electronics Engineering, or Electrical & Electronics Engineering.
2. Strong academic background in Power Electronics and Analog Electronics.
3. Basic understanding of electrical circuits, PCB schematics, and electronic components.

Preferred Skills

1. Proactive attitude toward learning, feedback, and assigned responsibilities.
2. Exposure to simulation and design tools such as LTspice, PSpice, MATLAB/Simulink, Altium Designer, KiCad, or similar EDA tools.
3. Familiarity with laboratory instruments such as Oscilloscope, Multimeter, Function Generator, DC Power Supply, and Power Analyzer.
4. Basic understanding of power electronic converter topologies such as Buck, Boost, Flyback, Forward, and LLC converters.
5. Knowledge of semiconductor devices (MOSFETs, IGBTs, diodes), magnetics, and fundamentals of analog/digital electronics.
6. Strong analytical and problem-solving skills with attention to detail and quality orientation.
7. Good communication and technical documentation skills, with the ability to work independently and collaborate in cross-functional teams.
8. Willingness to learn, adapt, and contribute effectively in a fast-paced R&D environment, with exposure to academic or personal projects in power electronics being an added advantage.